AP CALCULUS AB

Unit 1 • Lesson 1.7 practice • Answer key

Lesson 1.7 • Selecting Procedures for Determining Limits

5 multiple-choice questions and 1 free-response question • Answer key with explanations

Directions

Questions 1 to 5 are multiple choice: choose the best of the four answer choices. Question 6 is free response: show your work and justify your answers. Questions labeled **NO CALCULATOR** must be done without a calculator; questions labeled **CALCULATOR** require a graphing calculator. For questions with a graph, read the values you need from the grid.

Question 1

NO CALCULATOR

$\lim_{x \to 3} \dfrac{x+1}{(x-3)^2}$ is

(A) $+\infty$

(B) $-\infty$

(C) 0

(D) 4

Solution

Answer: (A). Substitution gives $\frac{4}{0}$. The denominator is positive on both sides, so the quotient increases without bound.

Question 2

NO CALCULATOR

$\lim_{x \to 0} \dfrac{x^2+5x}{x}$ is

(A) 0

(B) 5

(C) 1

(D) It does not exist.

Solution

Answer: (B). $\frac{0}{0}$: factor, $\dfrac{x(x+5)}{x} = x+5 \to 5$.

Question 3

NO CALCULATOR

$\lim_{x \to 1} \dfrac{x^2+1}{x+1}$ is

(A) 0

(B) 2

(C) 1

(D) It does not exist.

Solution

Answer: (C). Direct substitution works: $\frac{2}{2} = 1$.

Question 4

NO CALCULATOR

$\lim_{x\to 4} \dfrac{|x-4|}{x-4}$ is

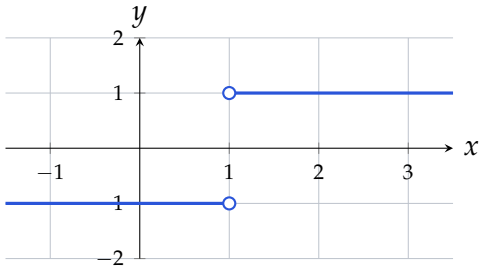
(A) 1
(B) -1
(C) 0
(D) It does not exist.

Solution
Answer: (D). The left-hand limit is -1 and the right-hand limit is 1.

Question 5 (graph)

NO CALCULATOR

The graph of $f(x) = \dfrac{|x-1|}{x-1}$ is shown. What is $\lim_{x\to 1} f(x)$?



(A) 1
(B) -1
(C) 0
(D) It does not exist.

Solution
Answer: (D). The left-hand limit is -1 and the right-hand limit is 1; they are not equal.

Question 6 (Free response)

NO CALCULATOR

For each limit, name the method you choose and evaluate the limit.

(a) $\lim_{x\to -1} \dfrac{x^2+2x+1}{x+1}$ (b) $\lim_{x\to 2^+} \dfrac{x}{2-x}$
(c) $\lim_{x\to 9} \dfrac{x-9}{\sqrt{x}-3}$ (d) $\lim_{x\to 0} \dfrac{\sin 2x}{\sin 3x}$

Solution
(a) $\frac{0}{0}$, factor: $\dfrac{(x+1)^2}{x+1} = x+1 \to 0$.
(b) $\frac{2}{0}$: the numerator is near $2 > 0$ and $2-x < 0$ for $x > 2$, so the limit is $-\infty$.
(c) $\frac{0}{0}$, factor $x-9 = (\sqrt{x}-3)(\sqrt{x}+3)$: the limit is $\sqrt{9}+3 = 6$.
(d) $\frac{0}{0}$, trig limit: $\dfrac{\sin 2x}{\sin 3x} = \dfrac{\frac{\sin 2x}{2x}\cdot 2x}{\frac{\sin 3x}{3x}\cdot 3x} \to \dfrac{1\cdot 2}{1\cdot 3} = \dfrac{2}{3}$.